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CLASS ACTIVITY

CSA0910 – JAVA PROGRAMMING

Generate a code and execute for the following

1. Multiple catch statements

```
import java.util.Scanner;

public class MultipleCatch {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        try {

            System.out.print("Enter number: ");

            int n = sc.nextInt();

            System.out.println(10 / n);

            int a[] = {10, 20, 30};

            System.out.println(a[n]);

        }

        catch (ArithmeticException e) {

            System.out.println("Arithmetic Exception");

        }

        catch (ArrayIndexOutOfBoundsException e) {

            System.out.println("Array Index Out Of Bounds Exception");

        }

        catch (Exception e) {

            System.out.println("Other Exception");

        }

    }

}
```

OUTPUT:

```
MultipleCatch.java
1 import java.util.Scanner;
2 public class MultipleCatch {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         try {
6             System.out.print("Enter number: ");
7             int n = sc.nextInt();
8             System.out.println(10 / n);
9             int a[] = {10, 20, 30};
10            System.out.println(a[n]);
11        }
12        catch (ArithmeticException e) {
13            System.out.println("Arithmetic Exception");
14        }
15        catch (ArrayIndexOutOfBoundsException e) {
16            System.out.println("Array Index Out Of Bounds Exception");
17        }
18        catch (Exception e) {
19            System.out.println("Other Exception");
20        }
21    }
22 }
```

Output

Enter number: 0  
Arithmetic Exception  
=== Code Execution Successful ===

## 2. StringIndexOutOfBoundsException

import java.util.Scanner;

public class StringIndex {

public static void main(String[] args) {

Scanner sc = new Scanner(System.in);

System.out.print("Enter a string: ");

String str = sc.nextLine();

try {

System.out.println("Character: " + str.charAt(10));

}

catch (StringIndexOutOfBoundsException e) {

System.out.println("String Index Out Of Bounds Exception");

}

}

}

OUTPUT:

```
StringIndex.java
1 import java.util.Scanner;
2 public class StringIndex {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         System.out.print("Enter a string: ");
6         String str = sc.nextLine();
7         try {
8             System.out.println("Character: " + str.charAt(10));
9         }
10        catch (StringIndexOutOfBoundsException e) {
11            System.out.println("String Index Out Of Bounds Exception");
12        }
13    }
14 }
```

Output

Enter a string: java  
String Index Out Of Bounds Exception  
=== Code Execution Successful ===

### 3. Finally

```
import java.util.Scanner;

public class FinallyDemo {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        try {

            System.out.print("Enter two numbers: ");

            int a = sc.nextInt();

            int b = sc.nextInt();

            System.out.println("Result = " + (a / b));

        }

        catch (ArithmeticException e) {

            System.out.println("Cannot divide by zero");

        }

        finally {

            System.out.println("Finally block executed");

        }

    }

}
```

OUTPUT:



```
FinallyDemo.java  Run  Output

1 import java.util.Scanner;
2 public class FinallyDemo {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         try {
6             System.out.print("Enter two numbers: ");
7             int a = sc.nextInt();
8             int b = sc.nextInt();
9             System.out.println("Result = " + (a / b));
10        }
11        catch (ArithmeticException e) {
12            System.out.println("Cannot divide by zero");
13        }
14        finally {
15            System.out.println("Finally block executed");
16        }
17    }
18 }
```

Enter two numbers: 10 2  
Result = 5  
Finally block executed  
=== Code Execution Successful ===

### 4. throw (to check person is eligible to poll the vote)

```
import java.util.Scanner;
```

```

public class Voting {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter age: ");

        int age = sc.nextInt();

        try {

            if (age < 18) {

                throw new Exception("Person is not eligible to vote");

            }

            System.out.println("Person is eligible to vote");

        }

        catch (Exception e) {

            System.out.println(e.getMessage());

        }

    }

}

```

OUTPUT:

| Voting.java                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Output                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <pre> 1 import java.util.Scanner; 2 public class Voting { 3     public static void main(String[] args) { 4         Scanner sc = new Scanner(System.in); 5         System.out.print("Enter age: "); 6         int age = sc.nextInt(); 7         try { 8             if (age &lt; 18) { 9                 throw new Exception("Person is not eligible to vote"); 10            } 11            System.out.println("Person is eligible to vote"); 12        } 13        catch (Exception e) { 14            System.out.println(e.getMessage()); 15        } 16    } 17 } </pre> | <pre> Enter age: 17 Person is not eligible to vote  === Code Execution Successful === </pre> |

## 5. Throws

```

import java.util.Scanner;

public class ThrowsDemo {

    static void divide(int a, int b) throws ArithmeticException {

        System.out.println("Result = " + (a / b));

    }

}

```

```

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter two numbers: ");

    int a = sc.nextInt();

    int b = sc.nextInt();

    try {

        divide(a, b);

    }

    catch (ArithmeticException e) {

        System.out.println("Cannot divide by zero");

    }

}
}

```

OUTPUT:

| ThrowsDemo.java                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Run | Output                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------------------------------------------------------------------------------------|
| <pre> 1 import java.util.Scanner; 2 public class ThrowsDemo { 3     static void divide(int a, int b) throws ArithmeticException { 4         System.out.println("Result = " + (a / b)); 5     } 6     public static void main(String[] args) { 7         Scanner sc = new Scanner(System.in); 8         System.out.print("Enter two numbers: "); 9         int a = sc.nextInt(); 10        int b = sc.nextInt(); 11        try { 12            divide(a, b); 13        } 14        catch (ArithmeticException e) { 15            System.out.println("Cannot divide by zero"); 16        } 17    } 18 } </pre> | Run | <pre> Enter two numbers: 24 6 Result = 4  === Code Execution Successful === </pre> |